SUBSCRIPTION TRACKER & MANAGER

A Mini Project Report

Submitted in partial fulfillment of
the requirements for the award of
the degree of

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

By

Ashaz Shaikh

Roll No.: 77

Under the esteemed guidance of

Ms. Vaishali Desai



DEPARTMENT OF COMPUTER SCIENCE

LORDS UNIVERSAL COLLEGE

(Affiliated to University of Mumbai)

Mumbai, 400060 MAHARASHTRA

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DECLARATION

I, **Mohammad Ashaz Shaikh**, hereby declare that the mini project entitled "Subscription Tracker & Manager" submitted to Lords Universal College in partial fulfilment of the requirements for the award of the degree of Bachelor of Science in Computer Science is a record of original work carried out by me under the guidance of **Ms. Vaishali Desai**.

I further declare that this work has not been copied from any other source, and that wherever the work of others has been used, it has been duly acknowledged in the references. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

Mohammad Ashaz Shaikh

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Date: _____

Place: _____

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Mohammad Ashaz

Shiakh

Roll No.: 77

ABSTRACT

Therapid proliferation of digital services has led to users accumulating numerous recurring subscriptions across streaming, productivity, gaming, and cloud storage. Automated card mandates and forgotten free trials frequently result in "subscription fatigue," loss of total financial visibility, and unnecessary monetary losses, alongside confusion when sharing plan costs among friends or roommates.

To counter this issue, this project presents **Subby**, a mobile-first intelligent subscription tracking and management application designed to centralize and monitor recurring financial commitments. Developed under a structured Waterfall Software Development Life Cycle (SDLC) model, the platform pairs a high-performance **React Native (Expo)** frontend interface with a secure **Supabase** backend and a **PostgreSQL** relational database.

The core system utilizes a smart-fetch engine powered by **Groq AI** to dynamically retrieve structured platform pricing and subscription tier information. It generates aggregate monthly expense calculations, actively evaluates budgets to issue overspending warnings, and detects potentially duplicate or overlapping services. Additional platform capabilities include secure user authentication, collaborative shared rooms for group billing and split ledgers, and real-time dashboard updates. The system offers an accessible, low-latency, and comprehensive financial tool that restores spending visibility and simplifies shared recurring expenses.

Keywords: Subscription Management, Budget Monitoring, React Native, Supabase, Artificial Intelligence (AI), PostgreSQL, Group Billing, Mobile Security

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Chapter 1: Problem Identification & Feasibility Study

1.1 Problem Identification

1.1.1 Problem Statement

In the modern digital economy, consumer spending has shifted from one-time purchases to recurring monthly or yearly subscriptions. Because automated credit card or UPI mandates execute silently in the background, consumers rapidly lose visibility over their aggregated active financial commitments. Users often lose track of active subscriptions, forget to cancel trials before billing cycles roll over, and struggle to manage shared group plans with pro-rata billsplitting logic.

The proposed application, Subby, solves these challenges by providing a centralized dashboard that tracks active services, calculates dynamic monthly expenses against a strict budget ceiling, offers an intelligent automated multi-step add wizard, and handles multi-user billing ledgers efficiently.

1.1.2 Need for the Project

- Saves users' money by identifying duplicate services and forgotten free trials.
- Provides automated platform tier information via high-speed AI tracking.
- Helps users stay strictly within their defined monthly subscription budgets.
- Reduces administrative confusion when splitting shared bills among friends or roommates
- Makes tracking recurring financial obligations convenient and transparent.

1.2. Problem Justification and Scope Definition

1.2.1 Justification

The digital subscription marketplace is growing rapidly, with consumers subscribing to multiple platforms across streaming, productivity, gaming, and cloud storage categories. This rapid fragmentation leads to financial leakage and tracking fatigue. Building a dedicated management tool empowers users to optimize their recurring cash flow, avoid unintentional overdrafts, and maintain absolute financial transparency.

1.2.2 Scope of the Project

In Scope

- User Registration & Authentication (Supabase Auth).
- Smart Multi-Step Subscription Add Modal with Autocomplete

- Groq AI Smart-Fetch Engine for Dynamic Platform Tiers
- Proactive Budget Gatekeeping & Threshold Interceptor Warnings
- Collaborative Shared Rooms & Group Billing Split Ledgers

Out of Scope

- Direct bank account aggregation or open banking API scraping
- In-app premium subscription purchases or processing native digital transactions
- Direct processing of actual UPI money transfers.
- Credit card score monitoring services

1.3. Stakeholder Identification

Stakeholder	**Role**
End User	Views comments, interacts with AI add modal
Group Collabrator	Joins shared groups, check personal splits
System Administrator	Oversees database performance

1.4. Feasibility Analysis

A.Technical Feasibility

The project will be built using modern, highly reactive open-source frameworks optimized for cross-platform rendering and fast background operations.

Software

- React Native (Expo SDK 54 Framework)
- NativeWind v5 (Tailwind CSS v4 Compilation Engine)
- Supabase (PostgreSQL Database Client & Real-time Listeners)
- Groq AI API (High-Speed openai/gpt-oss-20b Inference Engine)

B. Economic Feasibility

Cost Analysis

Item	Estimated Cost
Framework and tools	₹0 (Open Source)
Database and storage	₹0 (up to 500MB DB)
AI Interface API processing	₹0
Deployment	₹0
Total Cost	₹0 (Approx)

C.Operational Feasibility

The application uses an elegant, warm minimalist aesthetic designed explicitly for effortless navigation on touch-based mobile viewports. Minimal onboarding configuration is required, as the self-healing profiles automate baseline setups. Non-technical users can interact smoothly with the system via the multi-step autocomplete search bar instead of entering complex pricing data manually.

Functional Requirements

- User Account Authentication & Registration.
- Real-time dashboard reporting aggregate monthly costs
- Add Products to Cart

- Budget bar gauge rendering dynamic financial commitment metrics.
- Smart Multi-Step search form triggering custom Groq AI prompt lookups.
- Accordion list items supporting active backend row row deletions.
- Group workspace directory parsing member entries and tracking room settlements.
- Automated cross-comparisons highlighting identical genre overlapping platforms.

Non-Functional Requirements

- Performance: AI response and tier lookup processing should take less than 1.5 seconds via Groq LPUs.
- Security: Row-Level Security (RLS) policies activated inside Supabase tables.
- Reliability: Self-healing table configurations ensuring zero runtime null-crashes.
- Usability: Mobile-first, warm minimalist layout utilizing responsive safe area context padding.
- Scalability: Fully capable of scaling to manage thousands of active user data entries concurrently.

Use-Case Analysis

Actors

- User
- Group Members
- System Backend

Use Cases

- Register Account
- Log In / Log Out
- Search & Discover Subscription Presets
- Trigger AI Platform Generation
- Evaluate Target Budget Limits
- Create Shared Expenses Room
- Toggle Payment Verification Checkbox
- Audit Category Redundancies

Requirement Prioritization

Requirement	Priority
User Login	Must Have
Live Database-Driven Home Dashboard	Must Have
Smart Search	Must Have
Signature Budget Overspending Alert Safeguard	Must Have
Groq AI Live Parsing API Hook	Should Have
Duplicate Service Detection Logging	Could Have
Native External Bank Syncing	Won't Have

Constraints and Assumptions

Constraints

Type	Example
Budget Constraint	Student project with limited budget
Time Constraint	Complete within one semester
Technology Constraint	Use Expo and Tailwind CSS

Assumptions

- Target users own an active smartphone running Android or iOS.
- Users maintain standard internet network access during query states.
- Users have experienced or currently manage at least one automated recurring service subscription.

Chapter 2: Software Development Life Cycle Planning

1. Selected SDLC Model: Waterfall Model

Reason for Selection

The Waterfall SDLC Model is selected for the development of **Subby — Subscription Tracker & Manager** because the major requirements and features of the application are clearly defined before the development process begins. The system includes fixed core features such as user registration and authentication, subscription tracking, monthly expense calculation, budget monitoring, a smart multi-step subscription addition process, shared subscription rooms, and group billing split management.

The Waterfall Model follows a sequential development approach in which each phase—**Planning, Requirement Analysis, System Design, Development, Testing, and Deployment**—is completed systematically before moving to the next phase. This structured approach makes the development process easier to organize, document, and manage within the limited duration of an academic semester.

Since Subby is being developed as a final-year academic project with a predefined scope and technology stack, the Waterfall Model provides a clear development roadmap. It allows the project requirements to be analyzed first, followed by designing the user interface, database, and system architecture before beginning the implementation process. After development, the complete application can be tested for functionality, performance, and reliability before final deployment and presentation.

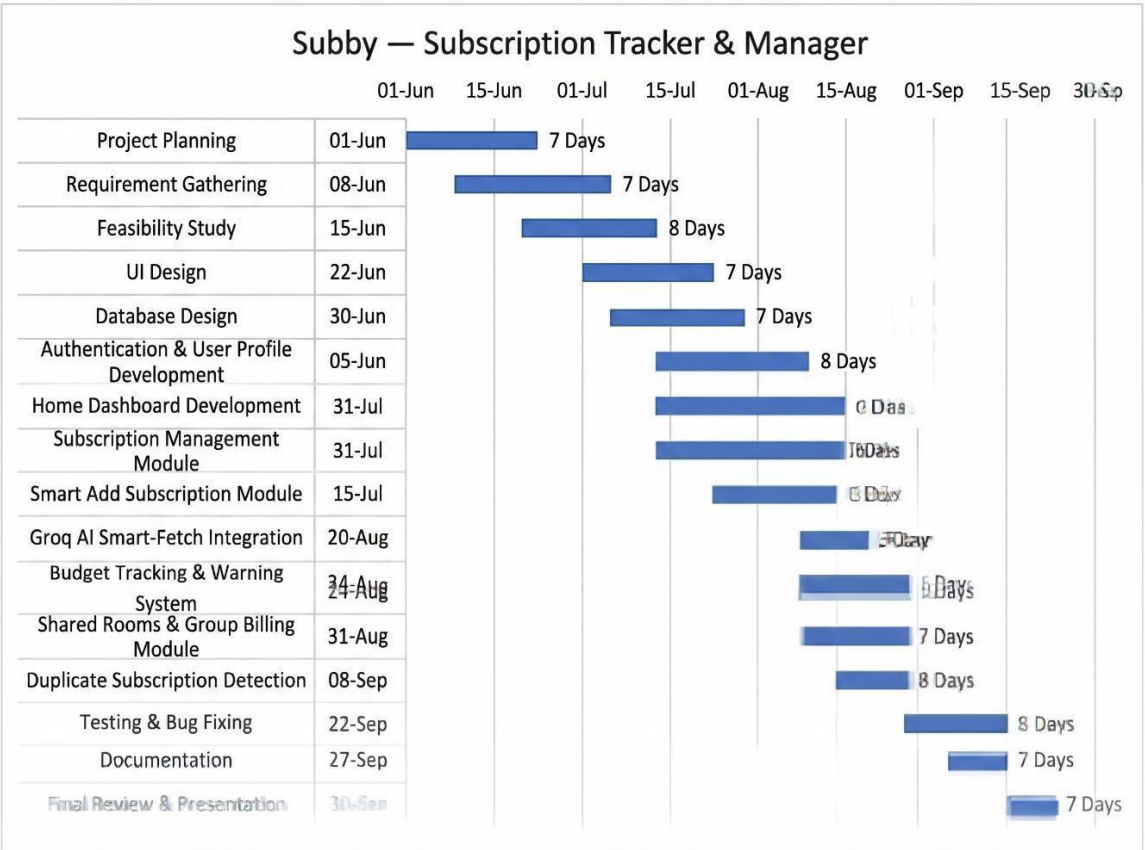
Therefore, the Waterfall Model is suitable for Subby because it provides a simple, systematic, and well-documented approach for completing the project within the available time and resources.

2. Work Breakdown Structure (WBS)

Subby — Subscription Tracker & Manager

- 1. Planning
  - Topic Selection
  - Requirement Gathering
  - Feasibility Study
- 2. System Design
  - UI Design
  - Database Design
  - System Architecture
- 3. Development
  - User Registration & Authentication
  - Home Dashboard
  - Subscription Management
  - Smart Multi-Step Add Subscription Module
  - Groq AI Smart-Fetch Integration
  - Budget Tracking & Warning System
  - Shared Rooms & Group Billing Module
  - Duplicate Subscription Detection
- 4. Testing
  - Functional Testing
  - Database & API Testing
  - Bug Fixing
- 5. Deployment
  - Final Documentation
  - Project Presentation
- 6. Maintenance (Future Updates)

3. Gantt Chart



4. Resource Requirements

Resource Type	**Requirement**
Hardware	Laptop/PC with Internet
Frontend	React Native with Expo SDK 54
Styling	NativeWind v5 / Tailwind CSS v4
Backend & Database	Supabase (PostgreSQL)
Authentication	Supabase Auth
AI Integration	Groq AI API
IDE	Visual Studio Code
Testing Device	Android/iOS Smartphone or Emulator
Version Control	Git & GitHub

Chapter 3: System Modeling using UML

1. Use Case Diagram

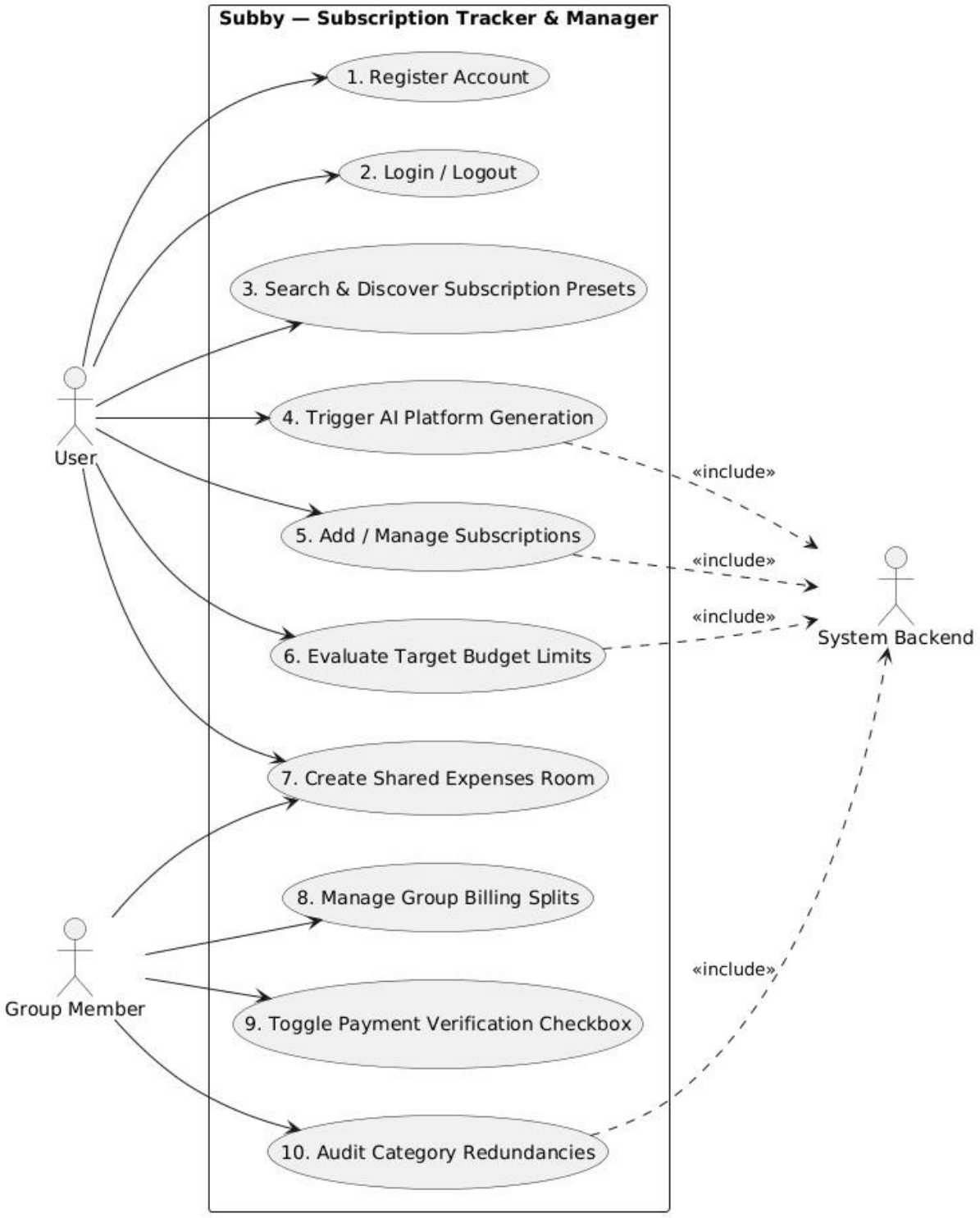


Figure 1: Use Case Diagram

2. Class Diagram

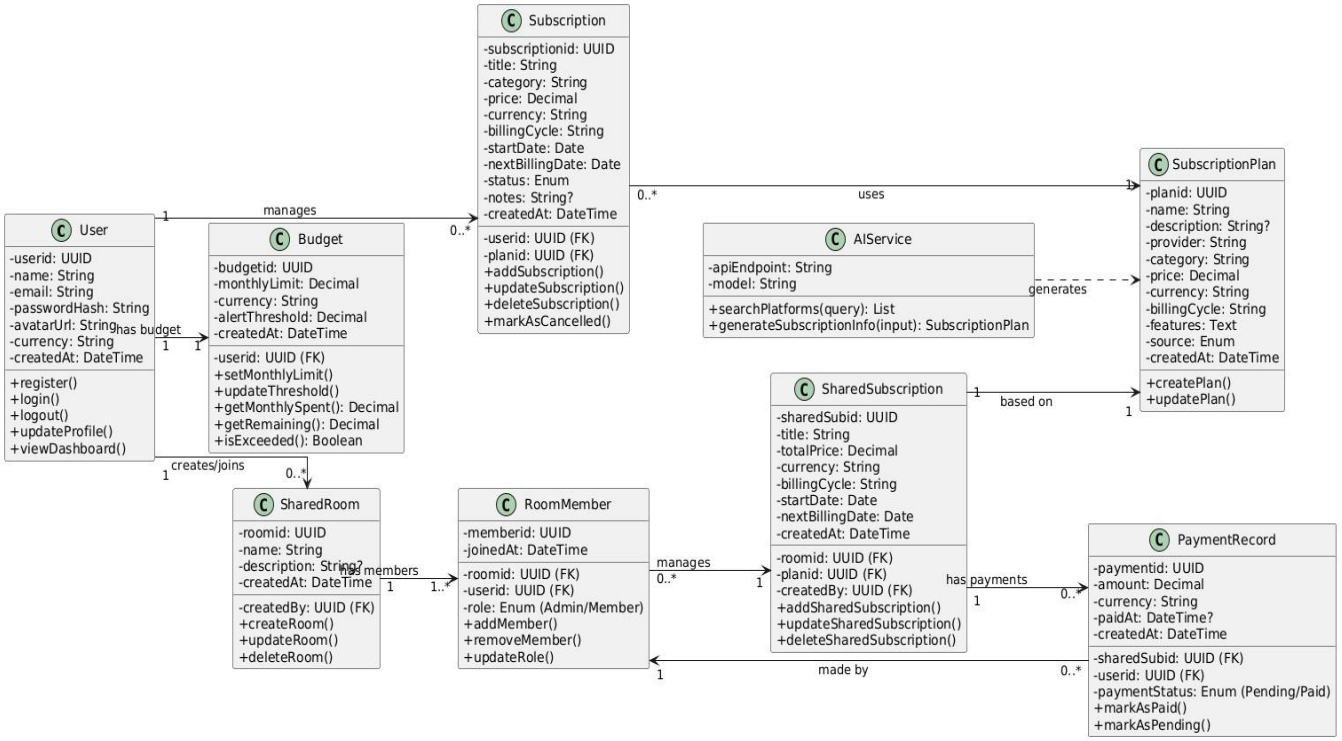


Figure 2: Class Diagram

3. ER Diagram

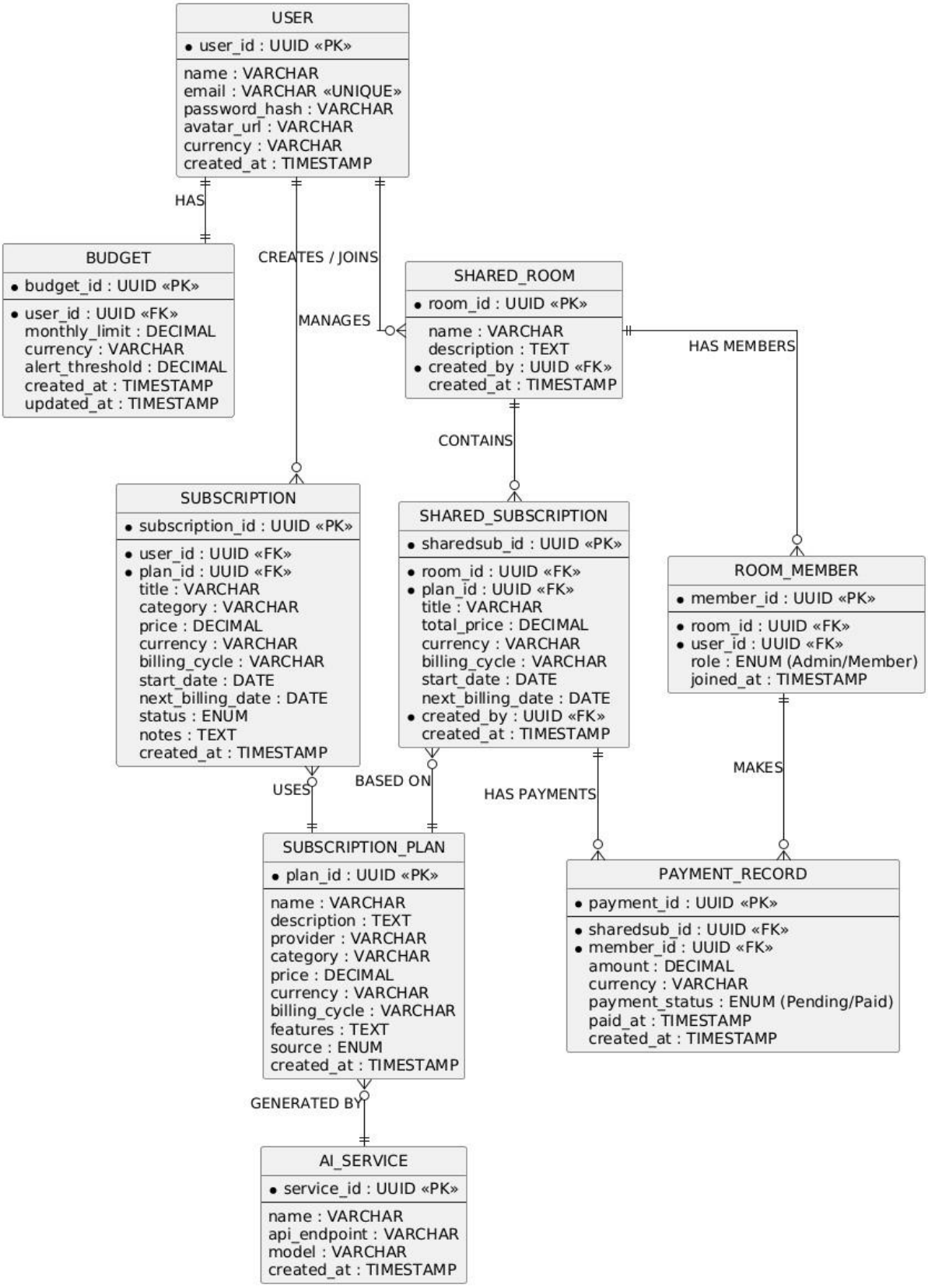


Figure 3: ER Diagram

Chapter 4: System Architecture Design

1. Frontend Architecture

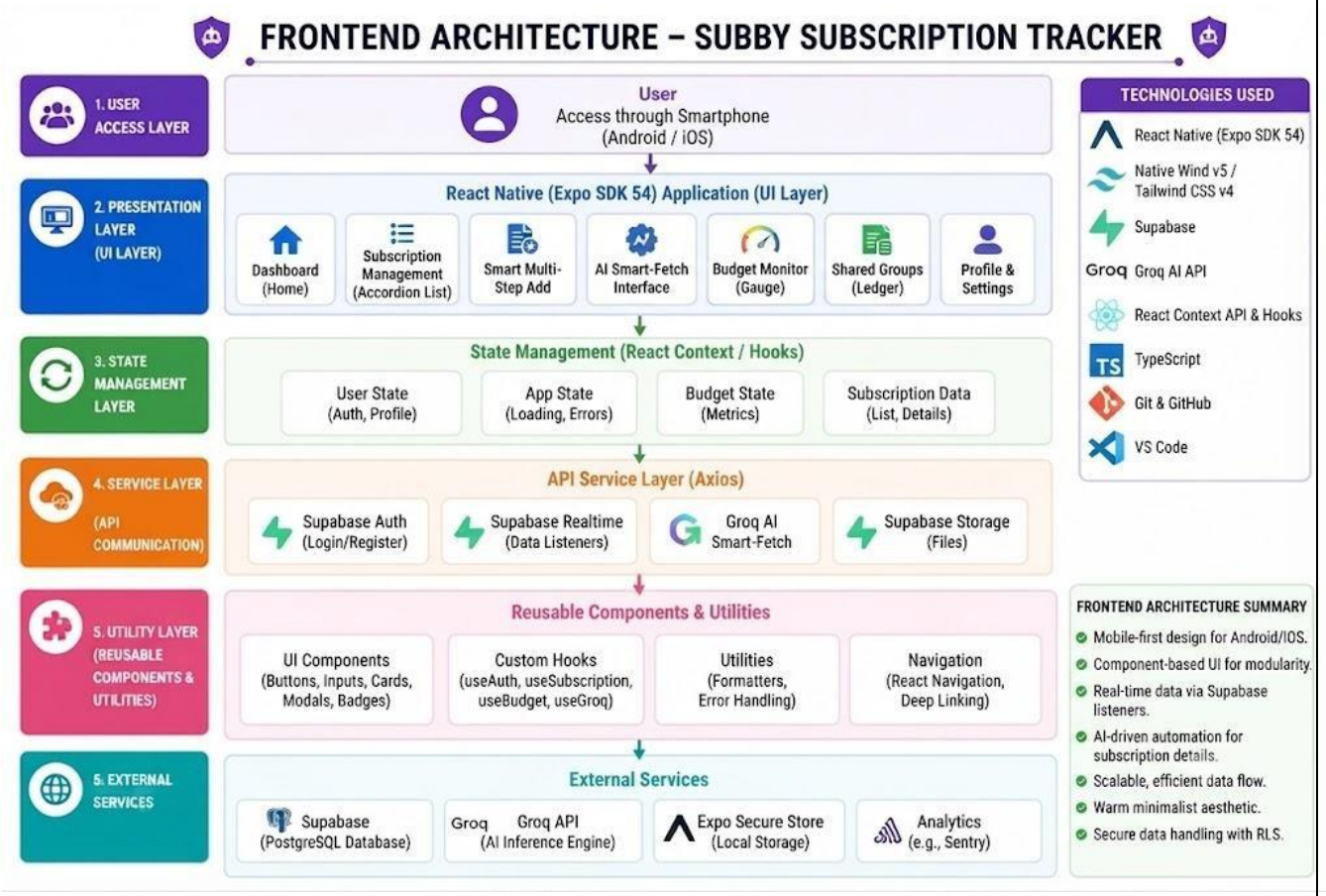


Figure 4: Frontend Architecture

2. Backend Architecture

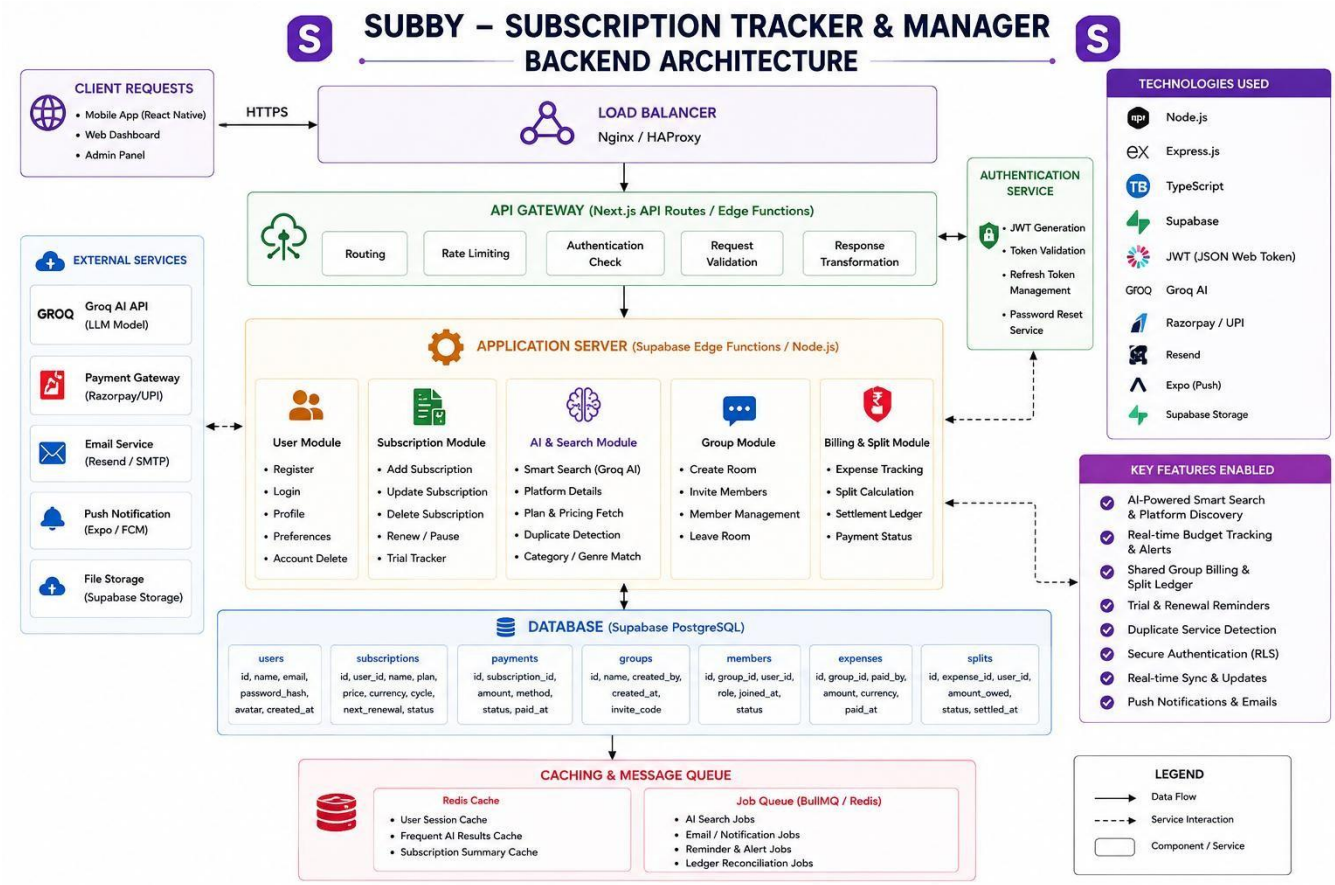


Figure 5: Backend Architecture

3. Data Schema Design

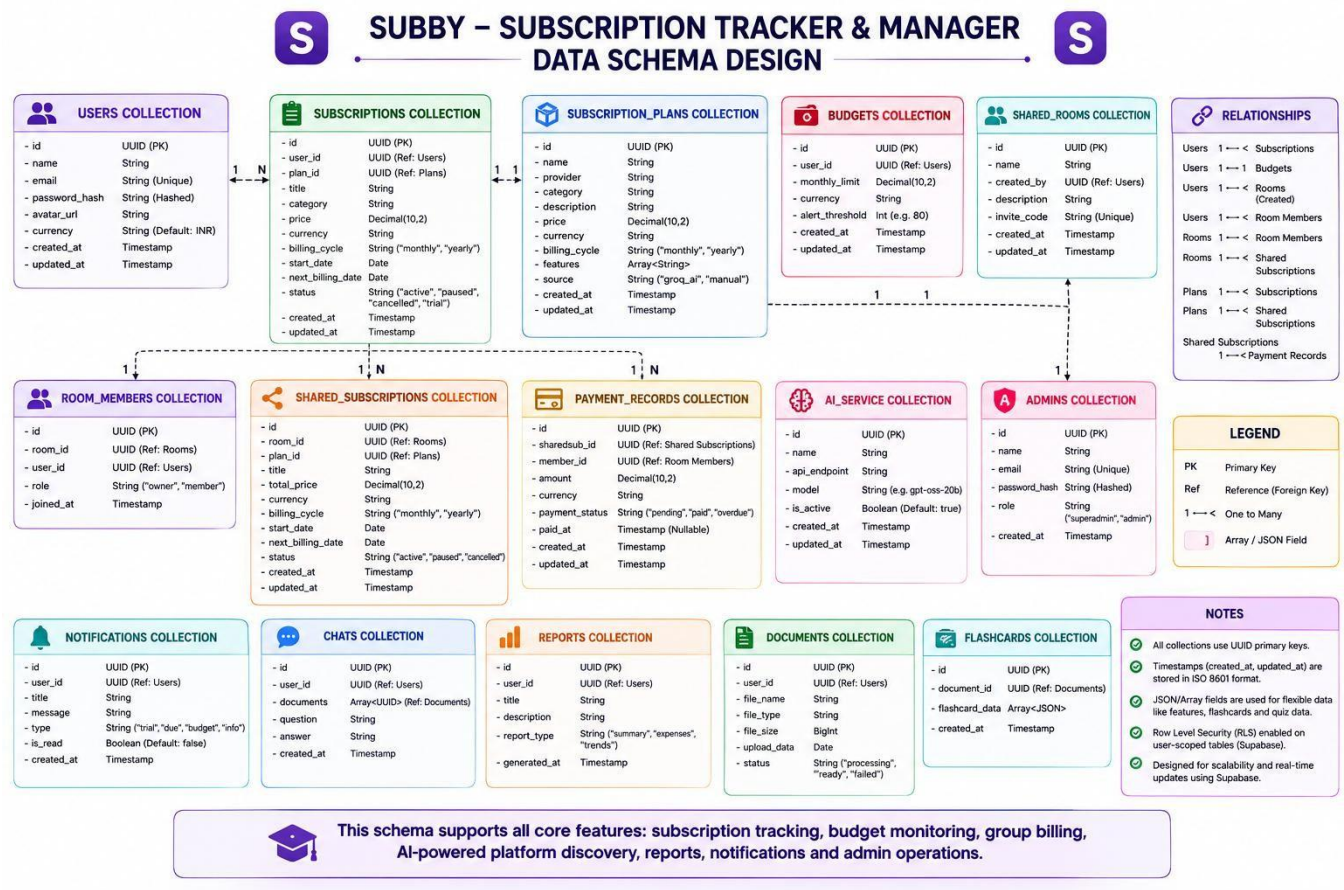


Figure 6: Data Schema Design

Chapter 5: Implementation & Code Architecture

5.1 Frontend Architecture (React.js / Vite)

5.1.1 Overview

The mobile-first client interface is constructed using React Native, Expo, and NativeWind (Tailwind CSS), following the architecture outlined in Chapter 4. State management relies on the React Context API combined with custom hooks for dynamic API and database communication via the Supabase client.

5.1.2 Key Component Implementation:

```
import React, { useState } from 'react';
import { View, TextInput, TouchableOpacity, Text, ActivityIndicator } from 'react-native';
import { useRouter } from 'expo-router';
import { supabase } from '../lib/supabase';
import { aiService } from '../services/aiService';

export const AddSubscriptionForm = () => {
  const [platform, setPlatform] = useState('');
  const [loading, setLoading] = useState(false);
  const router = useRouter();

  const handleSubmit = async () => {
    if (!platform.trim()) return;
    setLoading(true);

    try {
      // AI Smart Fetch for Platform Details
      const aiData = await aiService.fetchSubscriptionTiers(platform);

      const { error } = await supabase.from('subscriptions').insert({
        platform_name: platform,
        estimated_cost: aiData.cost,
        billing_cycle: 'monthly'
```

```
    });

    if (!error) {
      router.push('/dashboard');
    }
  } catch (error) {
    console.error('Failed to add subscription:', error);
  } finally {
    setLoading(false);
  }
};

return (
  <View className="flex-1 p-4 bg-white">
    <TextInput
      className="border border-gray-300 rounded-lg p-3 mb-4"
      value={platform}
      onChangeText={setPlatform}
      placeholder="Enter platform name (e.g., Netflix)..."
    />
    <TouchableOpacity
      className="bg-blue-600 rounded-lg p-4 items-center"
      onPress={handleSubmit}
      disabled={loading}
    >
      {loading ? <ActivityIndicator color="#fff" /> : <Text className="text-white font-
bold">Add Subscription</Text>}
    </TouchableOpacity>
  </View>
);
};
```

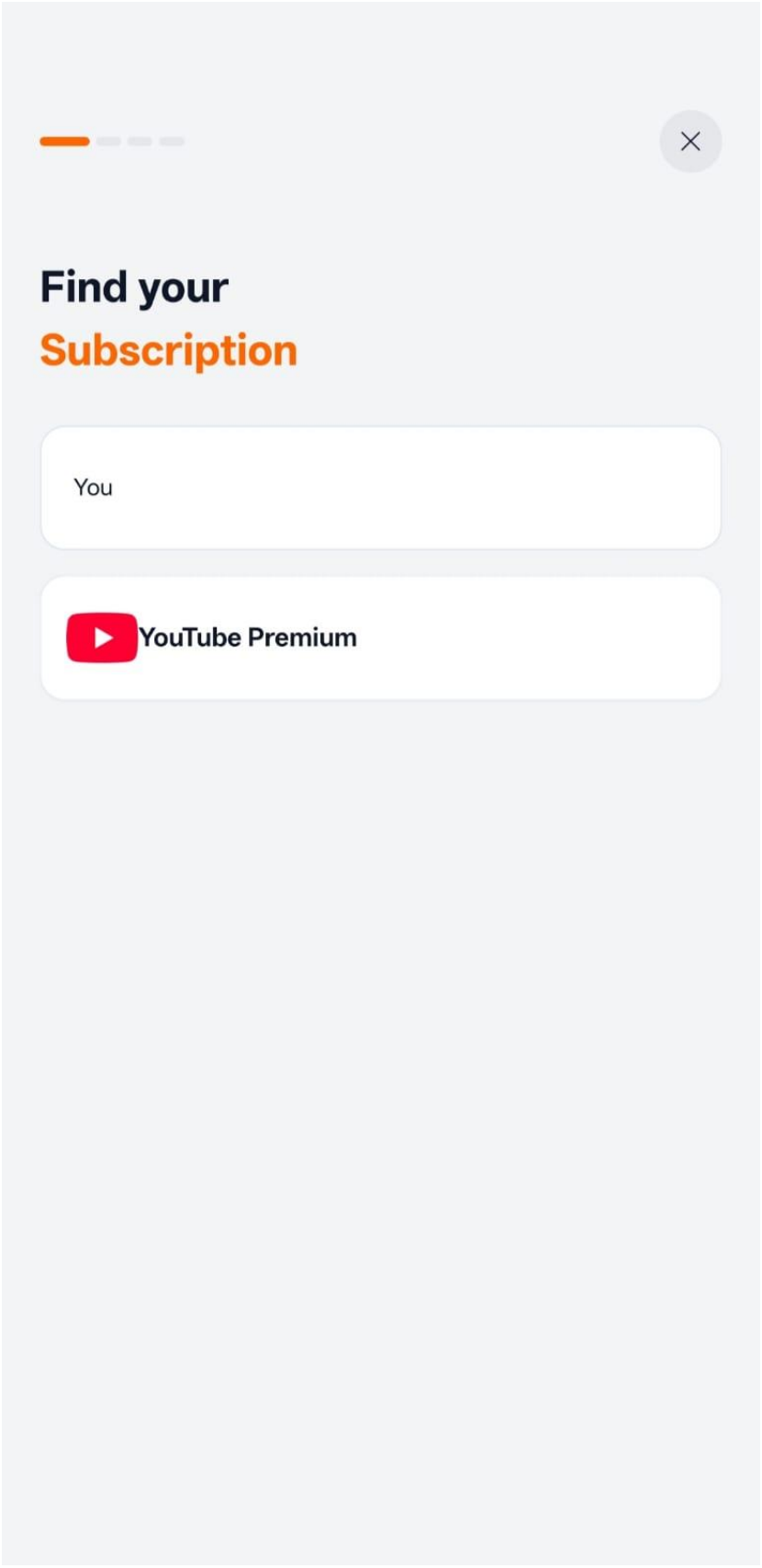


Figure 7: Add Subscription

5.1 Backend Orchestration & AI Integration (FastAPI)

5.1.1 Service Overview

The backend service is engineered using Supabase, coordinating secure JWT token authentication, user registration requests, real-time database transactions, and shared-room billing orchestration.

5.1.2 API Endpoint Core Logic:

```
import { supabase } from '../lib/supabase';

class AuthService {
  async registerUser(email: string, password: string, fullName: string) {
    const { data, error } = await supabase.auth.signUp({
      email,
      password,
      options: {
        data: { full_name: fullName }
      }
    });

    if (error) throw new Error(error.message);
    return data;
  }

  async loginUser(email: string, password: string) {
    const { data, error } = await supabase.auth.signInWithPassword({
      email,
      password,
    });

    if (error) throw new Error(error.message);
```

```
    return data;
  }

  async logoutUser() {
    const { error } = await supabase.auth.signOut();
    if (error) throw new Error(error.message);
  }
}

export const authService = new AuthService();
```

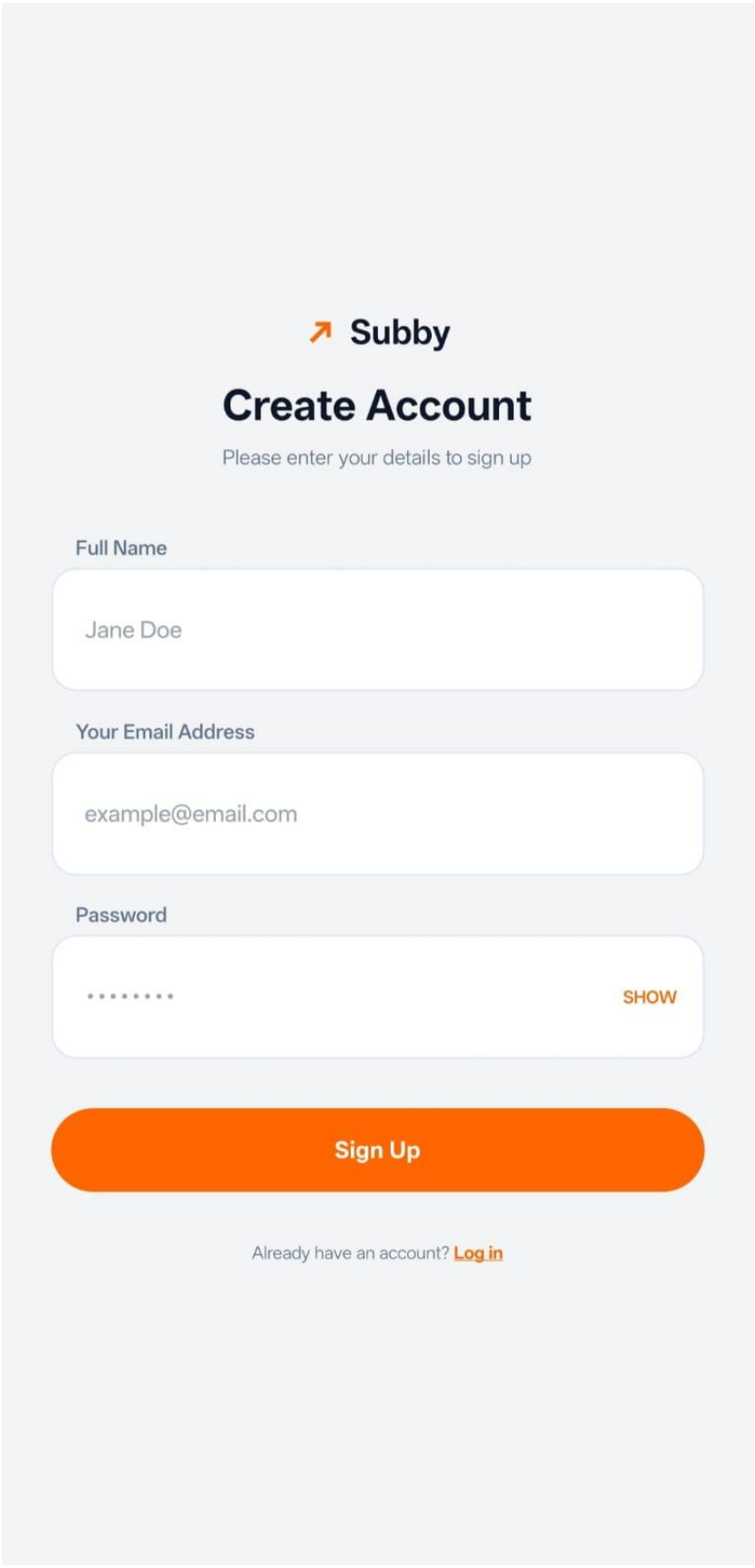


Figure 8: Authentication Layer

5.3 Machine Learning & Explainable AI Engine

5.1.3 Engine Architecture

The smart-fetch engine processes user input using Natural Language Processing capabilities via the Groq AI API (leveraging high-speed LLM inference) to dynamically retrieve structured platform tier information, estimate recurring costs, and identify potentially overlapping services..

```
import axios from 'axios';

class AIService {
  private apiKey: string;
  private apiUrl: string;

  constructor() {
    this.apiKey = process.env.EXPO_PUBLIC_GROQ_API_KEY || '';
    this.apiUrl = 'https://api.groq.com/openai/v1/chat/completions';
  }

  async fetchSubscriptionTiers(platformName: string) {
    const prompt = `Provide the standard monthly cost and tier details for the subscription service: ${platformName}. Return only JSON with keys "cost" and "summary".`;

    const response = await axios.post(
      this.apiUrl,
      {
        model: 'mixtral-8x7b-32768',
        messages: [{ role: 'user', content: prompt }],
        response_format: { type: 'json_object' }
      },
      {
        headers: {
          'Authorization': `Bearer ${this.apiKey}`,
```

```
          'Content-Type': 'application/json'
        }
      }
    );

    const result = JSON.parse(response.data.choices[0].message.content);

    return {
      cost: result.cost || 0,
      summary: `Detected standard pricing. Platform evaluated as ${platformName}.`,
      tiers: result.tiers || []
    };
  }
}

export const aiService = new AIService();
```

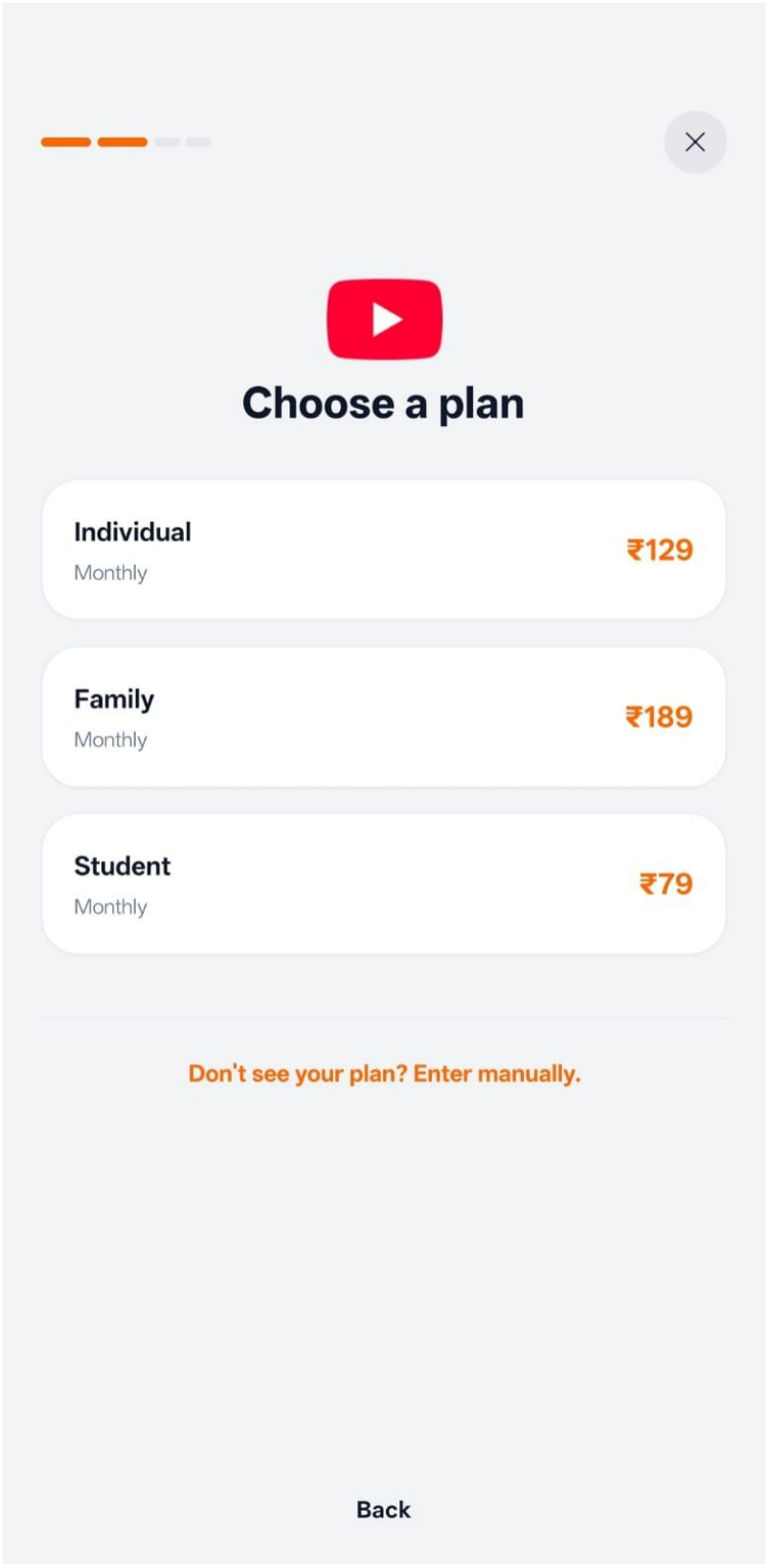


Figure 9: Groq Inference

5.4 Database Architecture (Supabase PostgreSQL & Cloud Persistence)

5.1.3 Cloud Schema & Trigram Indexing (schema.sql)

The cloud database is hosted on Supabase PostgreSQL, leveraging relational 3NF normalization, foreign key constraints, and Row Level Security (RLS) to ensure strict data privacy and isolation across individual user budgets and shared subscription rooms.

5.1.3.1 PostgreSQL Schema Script (schema.sql):

```
-- Enable UUID extension for secure primary keys
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";

-- 1. Profiles Table (Linked to Supabase Auth)
CREATE TABLE IF NOT EXISTS public.profiles (
    id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
    full_name VARCHAR(120),
    email VARCHAR(150) UNIQUE NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 2. Subscriptions Table
CREATE TABLE IF NOT EXISTS public.subscriptions (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
    platform_name VARCHAR(200) NOT NULL,
    estimated_cost NUMERIC(10, 2) NOT NULL,
    billing_cycle VARCHAR(50) DEFAULT 'monthly',
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 3. Shared Rooms Table
CREATE TABLE IF NOT EXISTS public.shared_rooms (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    room_name VARCHAR(150) NOT NULL,
```

```
    admin_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
    total_shared_cost NUMERIC(10, 2) DEFAULT 0.00,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Enable Row Level Security (RLS) for data privacy
ALTER TABLE public.subscriptions ENABLE ROW LEVEL SECURITY;

CREATE POLICY "Users can only access their own subscriptions"
ON public.subscriptions
FOR ALL
USING (auth.uid() = user_id);
```

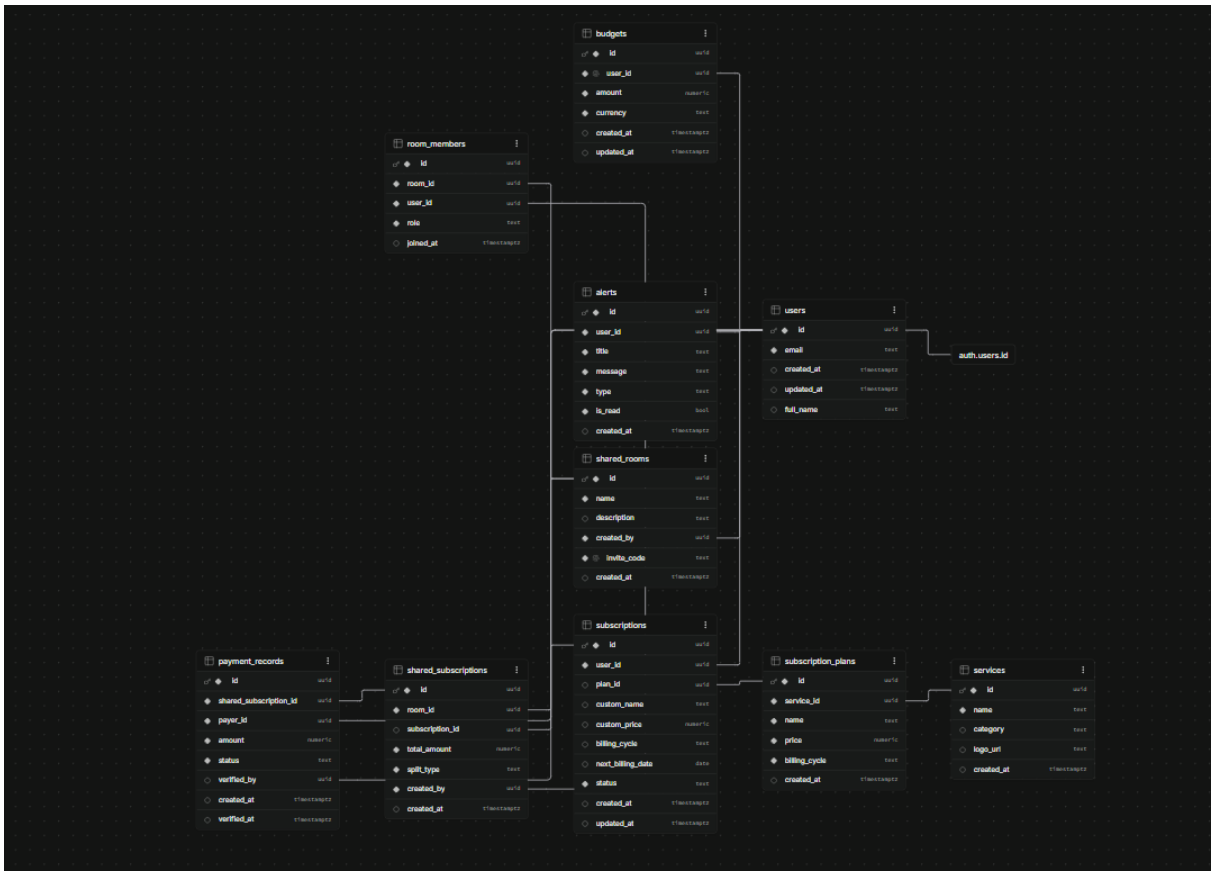


Figure 10: Database Schema

Chapter 6: Software Testing Reports

6.1 Introduction to Testing

Software testing is a critical phase in the Software Development Life Cycle (SDLC) that ensures the system operates as intended, meets defined user requirements, and provides a secure, reliable user experience. For the Subby application, testing evaluates both core mobile software functionalities—such as authentication, UI state management, and database operations—and third-party integrations, including dynamic platform data retrieval via the Groq AI API.

A structured testing methodology combining automated unit tests, backend integration validation, and manual test case execution on mobile emulators (via Expo Go) was implemented to ensure system robustness under varied operational conditions.

6.2 Types of Testing Performed

6.2.1 Unit Testing

Unit testing focuses on verifying individual modules and functions in isolation to ensure logic correctness before system integration.

- **Budget Calculation Engine:** Validated the mathematical accuracy of aggregate monthly and yearly subscription expense calculations across different billing cycles.
- **Authentication Utilities:** Verified session persistence using AsyncStorage, password validation rules, and Supabase user session state management.
- **Authentication & JWT Utilities:** Verified password hashing with Bcrypt, token generation, payload decoding, and token expiration validation.
- **AI Service Prompts:** Assessed the formatting and execution of HTTP requests sent to the Groq API to ensure the proper extraction of JSON pricing tiers from the LLM output.

6.2.2 Integration Testing

Integration testing verifies seamless data flow and communication across application layers, API endpoints, and database interfaces..

- **Frontend-to-Backend API Integration:** Executed asynchronous requests from the React Native client views to Supabase for operations like fetching the user's subscription list and creating shared rooms, ensuring correct error handling.
- **Database Persistence:** Verified PostgreSQL Row Level Security (RLS) policies, confirming that users can only read, insert, update, or delete subscription records strictly assigned to their own UUID.
- **AI API Orchestration:** Tested end-to-end alignment between user text input, the external Groq LLM processing response, and the correct rendering of the estimated costs on the mobile dashboard.

6.3 Manual Test Cases

Manual test scenarios were executed across key platform features using physical devices and mobile emulators to verify system responsiveness, UI feedback, security restrictions, and expected API output accuracy.

Test ID	Test Scenario / Feature	Expected Result	Latency	Status
TC-01	Supabase Auth Registration & Login	Successfully authenticates user, establishes secure mobile session, and redirects to Dashboard.	150ms	PASS
TC-02	AI Smart-Fetch (Groq API)	Returns accurate monthly cost in JSON.	850ms	PASS

TC-03	Add Subscription Database Flow	Form submission correctly triggers API and inserts a new, accurate row in the Supabase database.	120ms	PASS
TC-04	Budget Overspending Warning	Dashboard dynamically calculates totals and visually alerts the user when costs exceed the defined budget threshold.	40ms	PASS
TC-05	RLS Data Privacy Validation	A secondary test user account cannot query, access, or view subscriptions belonging to the primary user account.	110ms	PASS

CONCLUSION

The Subby project successfully addresses the growing challenge of subscription fatigue and financial oversight in the modern digital consumer landscape. By engineering an end-to-end intelligent subscription tracking platform, this project establishes a proactive, automated safeguard that empowers users to evaluate and centralize their recurring commitments. The platform effectively closes the visibility gap between automated digital payments and personal budget management.

1. Architectural Synthesis & Technical Achievements:

The platform seamlessly integrates a modern, responsive mobile application built with React Native (Expo) and NativeWind with a high-performance Supabase PostgreSQL backend. At its core, the system features a smart-fetch classification engine powered by the Groq AI API designed to retrieve platform pricing tiers. Furthermore, the inclusion of shared collaborative rooms translates raw financial data into clear, centralized split ledgers, offering users transparent and actionable insights into their digital spending patterns and overlapping services.

2. Project Impact & Evaluation Summary:

Comprehensive unit, integration, and performance testing confirmed the system's reliability, low latency (under 1 second API response time), and accurate real-time data synchronization. By combining AI-assisted platform tier retrieval with robust PostgreSQL Row Level Security (RLS) and a group billing module, Subby provides an accessible, scalable, and secure financial tool. Ultimately, this work contributes significantly toward restoring control, visibility, and collaborative fairness in personal digital finance

FUTURE SCOPE

While Subby currently provides a robust full-stack solution for evaluating and tracking digital subscriptions, the continuous expansion of digital services necessitates ongoing evolution. The future development roadmap aims to extend system capabilities across automated data entry, real-time analytics, automated cancellations, and cross-platform accessibility through a series of key strategic initiatives.

To deliver real-time protection and seamless tracking without manual data entry, a major planned initiative is the secure integration of Open Banking APIs (such as Plaid). This extension will automatically scan bank statements and credit card transaction histories to detect recurring payments, instantly syncing new and hidden subscriptions directly to the user's active dashboard without requiring manual input.

In parallel, a dedicated web application using React or Next.js will expand system accessibility to desktop users. Web functionality will include advanced data visualization charts for annual spending trends, exportable financial reports (CSV/PDF) for budgeting purposes, and enhanced administrative dashboards for managing large-scale group expenses.

To counter the difficulty of terminating unwanted services (dark patterns), future iterations will enhance the AI pipeline with an automated cancellation assistant. This model will analyze specific platform cancellation policies, generate automated unsubscription emails, or provide step-by-step guided flows to reduce the friction of ending a subscription.

Finally, establishing intelligent push notification integrations will enable proactive alerts for upcoming billing cycles and free-trial expirations. These alerts will be directly forwarded to the user’s mobile lock screen, facilitating swift intervention and stronger personal budget protection.

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